SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Security Incident Mapping (SIM) Web-Based System

Name	OLUWAGBEMI SEGUN IGNACIOUS
Matric No.	241201066
Department	Computer Science
Faculty	Computing
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Document Purpose

This Software Requirements Specification defines the requirements for the Security Incident Mapping (SIM) web-based system. The proposed system is intended to provide a simple platform through which members of the campus community can report security incidents and provide information that can support monitoring and response. The SRS explains the system scope, users, functional requirements, non-functional requirements, major components, data requirements, use cases, interfaces, constraints and acceptance criteria.

1. Introduction

1.1 Background

The SEN 202 assignment identifies insecurity and community crime as problems that can affect campus communities. It proposes a web platform called Security Incident Mapping (SIM) through which students or villa-mates can quickly report security crimes on campus, including incidents such as theft, physical assault and other serious security concerns. The assignment also requires the solution to consider students, staff and landlords because students may live in surrounding residences.

1.2 Purpose of the System

SIM will provide a centralized and easy-to-use interface for submitting security incident information. The system will organize incident reports into structured records so that authorized users can review, monitor and manage reports. The project is intentionally designed as a mini web interface rather than a complex production system, in accordance with the assignment requirement.

1.3 Scope

- Allow a user to submit a security incident report.
- Capture the incident category, description, date/time and location.
- Provide a mechanism for confidential or anonymous reporting where appropriate.
- Allow authorized personnel to review submitted incidents.
- Allow incident status to be monitored, for example: Reported, Under Review, Resolved.
- Provide basic incident information for monitoring and mapping purposes.
- Provide a simple interface suitable for a prototype/academic demonstration.

2. Overall Description

2.1 Product Perspective

SIM is a standalone web-based application for the academic project. A typical implementation consists of a browser-based front end, a back-end service for processing requests, and a data store for incident records. The components communicate through normal web requests. The design can later be extended with authentication, maps, notifications and analytics, but these features are not required for the initial mini-interface.

2.2 User Classes

User	Main Responsibilities
Student / Villa-mate	Submit and view the status of security incident reports where permitted.
Staff	Report incidents and provide relevant security information.
Landlord / Hostel Owner	Report incidents affecting surrounding student residences and provide relevant information.
Security Officer	Review reports, update incident status and support response/monitoring.
System Administrator	Manage system data, users/roles and general system configuration.

2.3 Operating Environment

- Modern web browser on a phone, tablet or computer.
- Internet/network connection for web access.
- Web server or local development server for the application.
- Database or structured storage for incident records.

3. Functional Requirements

ID	Requirement	Description
FR-01	Incident Reporting	The system shall allow an authorized or permitted user to submit a new security incident report.
FR-02	Incident Category	The system shall allow the reporter to select or enter an incident category.
FR-03	Incident Description	The system shall provide a field for a clear description of the incident.
FR-04	Date and Time	The system shall capture the date and time associated with the incident.
FR-05	Location	The system shall capture the location or area where the incident occurred.
FR-06	Confidentiality	The system should support confidential/anonymous reporting where this is enabled by the project design.
FR-07	Report Storage	The system shall store submitted reports as structured incident records.
FR-08	Report Review	Authorized security personnel shall be able to review submitted reports.
FR-09	Status Management	Authorized personnel shall be able to update the status of an incident.
FR-10	Incident Monitoring	The system shall provide a basic view for monitoring submitted incidents.
FR-11	Validation	The system shall validate required fields before accepting a report.
FR-12	Access Control	The system shall restrict administrative and security-management functions to authorized users.

4. Non-Functional Requirements

ID	Quality Attribute	Requirement
NFR-01	Usability	The interface should be simple, clear and easy to use on common mobile and desktop screens.
NFR-02	Performance	Normal form submission and page loading should complete within a reasonable time under ordinary project
NFR-03	Security	Access to administrative functions shall be restricted and sensitive incident information shall not be unnece
NFR-04	Privacy	The system should minimize personally identifying information and support confidential reporting where req
NFR-05	Reliability	A successfully submitted report should be stored correctly and should not be silently lost.
NFR-06	Maintainability	The application should use organized components and readable code so that changes can be made easily.
NFR-07	Compatibility	The interface should work with modern browsers on common devices.
NFR-08	Scalability	The design should allow future addition of more incident records, users and features.

5. System Use Cases

ID	Use Case	Actor(s)	Main Flow
UC-01	Report Security Incident	Student, Staff, Landlord	Enter incident details → validate information → submit → store report → show
UC-02	Review Incident	Security Officer	Open incident list → select report → view submitted information.
UC-03	Update Incident Status	Security Officer	Open report → choose status → save update → system records the new stat
UC-04	Monitor Incidents	Security Officer, Administrator	View incident records and basic status information for monitoring.
UC-05	Manage System	Administrator	Manage permitted users, records and system configuration.

6. Major System Components

- **Presentation Layer:** Web pages/forms used by reporters, security personnel and administrators.
- **Application Layer:** Processes submissions, validation, permissions, incident status and business rules.
- **Data Layer:** Stores incident details, users/roles and status information.
- **Reporting/Monitoring Component:** Displays submitted incidents and their current status to authorized personnel.

7. Data Requirements

Data Item	Example/Description
Incident ID	Unique identifier for each report.
Incident Category	Theft, assault, vandalism, intrusion, etc.
Description	Narrative description of what was reported.
Location	Campus area or nearby residence/location.
Date/Time	When the incident occurred or was reported.
Reporter Type	Student, staff or landlord/hostel owner.
Report Status	Reported, Under Review, Resolved, or similar project-defined states.
Created/Updated Time	System timestamps for record tracking.

8. User Interface Requirements

- A clear landing page explaining the purpose of SIM.
- A simple incident-reporting form with clearly labelled fields.
- Visible validation messages when required information is missing.
- A confirmation message after successful submission.
- A protected dashboard/list for authorized personnel to review reports.
- Status indicators for tracking the progress of reports.
- Responsive layout so the mini-interface remains usable on mobile and desktop screens.

9. Security and Privacy Requirements

- Only authorized users should access security-management functions.
- Incident information should be handled as sensitive information.
- The system should avoid collecting unnecessary personal information.
- If anonymous reporting is enabled, the interface should not unnecessarily reveal the reporter's identity.
- Administrative actions should be restricted and, in a fuller implementation, logged for accountability.
- Input validation should be applied to reduce invalid or malicious data submissions.

10. Constraints and Assumptions

- The assignment requires a mini web interface and discourages unnecessary complexity.
- Users are assumed to have access to a web browser and network connection.
- The system is an academic prototype and is not presented as a replacement for emergency services or law enforcement.
- The accuracy of reports depends on information supplied by users.
- Advanced real-time mapping, SMS alerts and large-scale analytics are outside the minimum project scope unless implemented as extensions.

11. Traceability to SEN 202 Assignment

The assignment requires a questionnaire using Google Forms, Ishikawa problem analysis using Draw.io, a mini web interface, use-case diagram(s), an SRS containing functional and non-functional requirements, and storage of the application and related documentation in GitHub. This SRS directly documents the requirements and interactions of the proposed SIM application and is intended to be stored in the same GitHub repository as the other project materials.

Assignment Requirement	SRS Coverage
Questionnaire for students, staff and landlords	User classes, data requirements and system requirements.
Ishikawa analysis	Problem/background and scope; separate Ishikawa diagram stored in repository.
Mini web interface	Scope, interface requirements and major system components.
Use-case diagram(s)	User classes and detailed use-case descriptions.
SRS with functional/non-functional requirements	Sections 3 and 4 of this document.

Assignment Requirement	SRS Coverage
GitHub repository	Document is intended for inclusion in the project repository.

12. Acceptance Criteria

- A user can open the SIM interface and understand its purpose.
- A user can complete and submit an incident report using the required fields.
- The system validates required information before submission.
- A submitted incident is stored and can be retrieved by an authorized reviewer.
- An authorized security officer can review an incident and update its status.
- The interface works on common mobile and desktop browsers.
- The repository contains the SRS and other required project documentation.

13. Conclusion

The SIM system provides a focused software solution for improving the organization of campus security incident reports. The requirements in this SRS define a manageable academic prototype that supports reporting, review, status tracking and basic monitoring while considering privacy, security and usability. The design can be extended in future with richer mapping, notifications, analytics and integration with institutional security processes.

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